Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If the sum of the squares of the distance of the point (x, y, z) from the points $(a, 0, 0)$ and $(-a, 0, 0)$ is $2c^2$, then which one of the following is correct?

(A) $x^2 + a^2 = 2c^2 - y^2 - z^2$

(B) $x^2 + a^2 = c^2 - y^2 - z^2$

(C) $x^2 - a^2 = c^2 - y^2 - z^2$

(D) $x^2 + a^2 = c^2 + y^2 + z^2$

Q.2

The co-ordinates of the point in which the line joining the points $(3, 5, -7)$ and $(-2, 1, 8)$ is intersected by the plane yz are given by

(A) $\left(0, \frac{13}{5}, 2\right)$

(B) $\left(0, -\frac{13}{5}, -2\right)$

(C) $\left(0, -\frac{13}{5}, \frac{2}{5}\right)$

(D) $\left(0, \frac{13}{5}, \frac{2}{5}\right)$

Q.3

If $P(3, 2, -4)$, $Q(5, 4, -6)$ and $R(9, 8, -10)$ are collinear, then R divides PQ in the ratio

(A) $3 : 2$ internally

(B) $3 : 2$ externally

(C) $2 : 1$ internally

(D) $2 : 1$ externally

Q.4

Points $(-2, 4, 7)$, $(3, -6, -8)$ and $(1, -2, -2)$ are

(A) Collinear

(B) Vertices of an equilateral triangle

(C) Vertices of an isosceles triangle

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(D) None of these

Q.5

Which of the following statement is true?

(A) The point $\mathrm{A}(0,-1), \mathrm{B}(2,1), \mathrm{C}(0,3)$ and $\mathrm{D}(-2,1)$ are vertices of a rhombus.

(B) The points $\mathrm{A}(-4,-1), \mathrm{B}(-2,-4), \mathrm{C}(4,0)$ and $\mathrm{D}(2,3)$ are vertices of a square.

(C) The points $\mathrm{A}(-2,-1), \mathrm{B}(1,0), \mathrm{C}(4,3)$ and $\mathrm{D}(1,2)$ are vertices of a parallelogram.

(D) None of these

Q.6

Find the equation of set points P such that $PA^2 + PB^2 = 2K^2$, where A and B are the points $(3,4,5)$ and $(-1,3,-7)$, respectively :

(A) $K^2 - 109$

(B) $2K^2 - 109$

(C) $3K^2 - 109$

(D) $4K^2 - 10$

Q.7

ABC is a triangle and AD is the median. If the coordinates of A are $(4,7,-8)$ and the coordinates of centroid of the triangle ABC are $(1,1,1)$, what are the coordinates of D ?

(A) $\left(-\frac{1}{2}, 2, 11\right)$

(B) $\left(-\frac{1}{2}, -2, \frac{11}{2}\right)$

(C) $(-1,2,11)$

(D) $(-5,-11,19)$

Q.8

The equation of locus of a point whose distance from the y -axis is equal to its distance from the point $(2,1,-1)$ is

(A) $x^2 + y^2 + z^2 = 6$

(B) $x^2 - 4x + 2z + 6 = 0$

(C) $y^2 - 2y - 4x + 2z + 6 = 0$

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(D) $x^2 + y^2 - z^2 = 0$

Q.9

The ratio in which the join of points $(1, -2, 3)$ and $(4, 2, -1)$ is divided by XOY plane is

(A) $1 : 3$

(B) $3 : 1$

(C) $-1 : 3$

(D) None of these

Q.10

Points $(1, 1, 1), (-2, 4, 1), (-1, 5, 5)$ and $(2, 2, 5)$ are the vertices of a

(A) Rectangle

(B) Square

(C) Parallelogram

(D) Trapezium

Q.11

If x co-ordinates of a point P of line joining the points $\mathrm{Q}(2, 2, 1)$ and $\mathrm{R}(5, 2, -2)$ is 4, then the z - coordinates of P is

(A) -2

(B) -1

(C) 1

(D) 2

Q.12

A parallelopiped is formed by planes drawn through the points $(2, 3, 5)$ and $(5, 9, 7)$ parallel to the coordinate planes. The length of the diagonal of the parallelopiped is

(A) 8

(B) 4

(C) 7

(D) 11

Q.13

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The co-ordinates of the points A and B are $(2, 3, 4)$ and $(-2, 5, -4)$ respectively. If a point P moves so that $PA^2 - PB^2 = k$ where k is a constant, then the locus of P is

(A) $-8x + 4y - 16z + 16 = k$

(B) $-8x - 4y - 16z - 16 = k$

(C) $-8x + 4y - 16z - 16 = k$

(D) None of these

Q.14

$\mathrm{P(a, b, c); Q(a + 2,\ b + 2, c - 2)}$ and
$\mathrm{R(a + 6,\ b + 6, c - 6)}$ are collinear.
Consider the following statements :
1. R divides PQ internally in the ratio $3 : 2$
2. R divides PQ externally in the ratio $3 : 2$
3. Q divides PR internally in the ratio $1 : 2$ Which of the statements given above is/are correct?

(A) 1 only

(B) 2 only

(C) 1 and 3

(D) 2 and 3

Q.15

What is the locus of a point which is equidistant from the points $(1, 2, 3)$ and $(3, 2, -1)$?

(A) $x + z = 0$

(B) $x - 3z = 0$

(C) $x - z = 0$

(D) $x - 2z = 0$

Q.16

Which of the following statements is correct:

(A) Two non collinear vectors are always linearly dependent

(B) Two parallel non zero vectors are always linearly independent

(C) Any four vectors in $3 - d$ space are always linearly dependent

(D) None of these

Q.17

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The ratio in which the line joining $(2, 4, 5), (3, 5, -4)$ is divided by the yz plane, is

(A) $2 : 3$

(B) $3 : 2$

(C) $-2 : 3$

(D) $4 : -3$

Q.18

The points $(5, 2, 4), (6, -1, 2)$ and $(8, -7, \mathrm{k})$ are collinear if k is equal to

(A) -2

(B) 2

(C) 3

(D) -1

Q.19

What are coordinates of the point equidistant from the points $(\mathrm{a}, 0, 0), (0, \mathrm{a}, 0), (0, 0, \mathrm{a})$ and $(0, 0, 0)$?

(A) $\left(\frac{a}{3}, \frac{a}{3}, \frac{a}{3}\right)$

(B) $\left(\frac{\mathrm{a}}{2}, \frac{\mathrm{a}}{2}, \frac{\mathrm{a}}{2}\right)$

(C) $(\mathrm{a}, \mathrm{a}, \mathrm{a})$

(D) $(2\mathrm{a}, 2\mathrm{a}, 2\mathrm{a})$

Q.20

The xy -plane divides the line joining the points $(-1, 3, 4)(2, -5, 6)$

(A) internally in the ratio $2 : 3$

(B) externally in the ratio $2 : 3$

(C) internally in the ratio $3 : 2$

(D) externally in the ratio $3 : 2$

Answers & Solutions

Q.1 Answer:

$x^2 + a^2 = c^2 - y^2 - z^2$

Solution:

Let the point be $\mathrm{P}(\mathrm{x}, \mathrm{y}, \mathrm{z})$ **and two points,** $(a, 0, 0)$ **and** $(-a, 0, 0)$ **be** A **and** B **As given in the problem,** $\mathrm{PA}^2 + \mathrm{PB}^2 = 2\mathrm{c}^2$

so, $(x + a)^2 + (y - 0)^2 + (z - 0)^2$

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$+(x-a)^2+(y-0)^2+(z-0)^2=2c^2$
or $(x+a)^2+y^2+z^2+(x-a)^2+y^2+z^2=2c^2$
$\Rightarrow x^2+2a+a^2+y^2+z^2+x^2-2a+a^2+y^2+z^2=2c^2$
$\Rightarrow 2\left(\mathrm{x}^2+\mathrm{y}^2+\mathrm{z}^2+\mathrm{a}^2\right)=2\mathrm{c}^2$
$\Rightarrow x^2+y^2+z^2+a^2=c^2 \Rightarrow x^2+a^2=c^2-y^2-z^2$

Q.2 Answer:

$\left(0, \frac{13}{5}, 2\right)$

Solution:

Ratio $=-\left(\frac{3}{-2}\right)=\frac{3}{2}$
Required co-ordinates of the points are, $\left[\frac{6-6}{5}, \frac{10+3}{5}, \frac{-14+24}{5}\right]=\left(0, \frac{13}{5}, 2\right)$

Q.3 Answer:

$3:2$ externally

Solution:

Suppose R divides PQ **in the ratio** $\lambda:1$**. Then, the coordinates of** R **are**
$\left(\frac{5\lambda+3}{\lambda+1}, \frac{4\lambda+2}{\lambda+1}, \frac{-6\lambda-4}{\lambda+1}\right)$
But, the coordinates of R **are given as** $(9,8,-10)$
$\therefore \quad \frac{5\lambda+3}{\lambda+1}=9, \frac{4\lambda+2}{\lambda+1}=8$ **and**
$\frac{-6\lambda-4}{\lambda+1}=-10 \Rightarrow \lambda=-\frac{3}{2}$
Hence, R **divides** PQ **externally in the ratio** $3:2$

Q.4 Answer:

Collinear

Solution:

Here, $\frac{(3-(-2))}{1-3}=\frac{-6-4}{-2-(-6)}=\frac{-8-7}{-2-(-8)}$
$\Rightarrow -\frac{5}{2}=-\frac{5}{2}=-\frac{5}{2}$**. Obviously, points are collinear**

Q.5 Answer:

The points $\mathrm{A}(-2,-1), \mathrm{B}(1,0), \mathrm{C}(4,3)$ and $\mathrm{D}(1,2)$ are vertices of a parallelogram.

Solution:

Here $\mathrm{A}(0,-1), \mathrm{B}(2,1), \mathrm{C}(0,3), \mathrm{D}(-2,1)$**. For a rhombus all four sides are equal but the diagonal are not equal, we see** $AC=\sqrt{0+4^2}=4$
$\mathrm{BD}=\sqrt{4^2-0}=4$
Since diagonals are equals therefore it is a square, not rhombus
Here $\quad \mathrm{AB}=\sqrt{2^2+(-3)^2}=\sqrt{13}, \quad \mathrm{BC}$

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$= \sqrt{6^2 + 4^2} = \sqrt{52}$

Since $AB \neq BC$ therefore it is not square.

In this case mid point of AC is $\left(\frac{4-2}{2}, \frac{3-1}{2}\right)$ or $(1, 1)$

Also mid-point of diagonal BD $\left(\frac{1+1}{2}, \frac{0+2}{2}\right)$ or $(1, 1)$

Hence the points are vertices of a parallelogram.

Q.6 Answer:

$2K^2 - 109$

Solution:

Let the coordinates of point P be (x, y, z). Here, $PA^2 = (x-3)^2 + (y-4)^2 + (z-5)^2$

$PB^2 = (x+1)^2 + (y-3)^2 + (z+7)^2$

By the given condition $PA^2 + PB^2 = 2K^2$ We have $(x-3)^2 + (y-4)^2 + (z-5)^2 +$

$(x+1)^2$

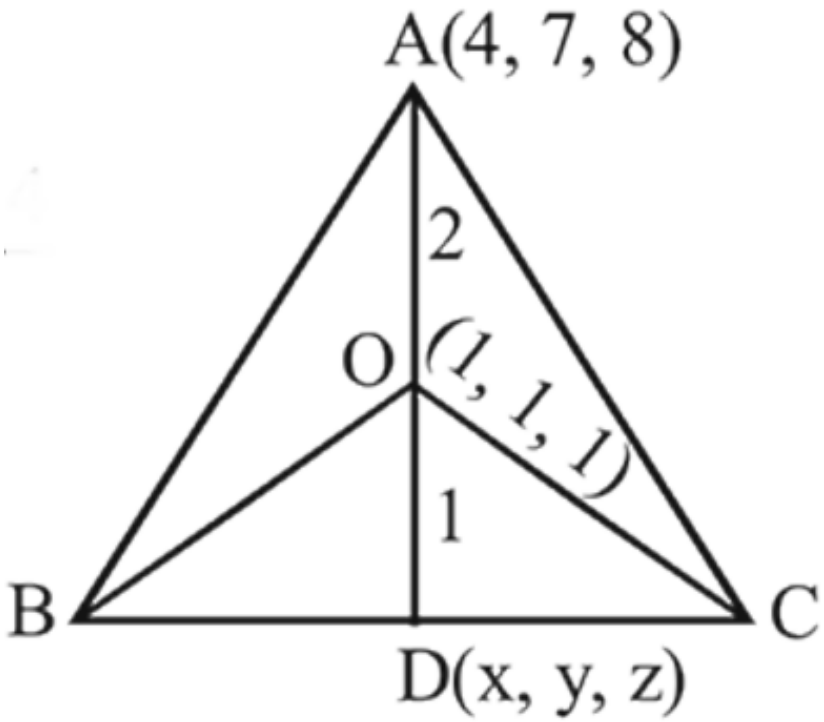
$+(y-3)^2 + (z+7)^2 = 2K^2$

i.e. $2x^2 + 2y^2 + 2z^2 - 4x - 14y + 4z = 2K^2 - 109$

Q.7 Answer:

$\left(-\frac{1}{2}, -2, \frac{11}{2}\right)$

Solution:



Let coordinates of D be (x, y, z) Co-ordinates of centroid is $(1, 1, 1)$, and of A, is $(4, 7, 8)$

Centroid divides median is $2 : 1$ ratio

So, $\frac{AO}{OD} = 2 : 1$

For x :

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$\Rightarrow x = -1/2$

For y : $1 = \frac{2y+1\times7}{1+2}$

$\Rightarrow y = -2$

For z : $1 = \frac{2\times z+1\times-8}{3} \Rightarrow z = +11/2$

$\therefore$ **Co-ordinates of** D **are** $(-1/2, -2, 11/2)$

Q.8 Answer:

$y^2 - 2y - 4x + 2z + 6 = 0$

Solution:

The variable point is $P(x, y, z)$.

Its distance from the y **-axis** $= \sqrt{x^2 + z^2}$

Its distance from $(2, 1, -1) = \sqrt{(x-2)^2 + (y-1)^2 + (z+1)^2}$

Given:

$\sqrt{x^2 + z^2} = \sqrt{(x-2)^2 + (y-1)^2 + (z+1)^2}$

$\Rightarrow y^2 - 2y - 4x + 2z + 6 = 0$

Q.9 Answer:

$3 : 1$

Solution:

Let $A(1, -2, 3)$ **and** $B(4, 2, -1)$. **Let the plane** XOY **meet the line** AB **in the point** C **such that** C **divides** AB **in the ratio** $k : 1$, **then** $C \equiv \left(\frac{4k+1}{k+1}, \frac{2k-2}{k+1}, \frac{-k+3}{k+1}\right)$. **Since C lies on the plane** XOY **i.e. the plane** $z = 0$, **therefore,** $\frac{-k+3}{k+1} = 0 \Rightarrow k = 3$

Q.10 Answer:

Square

Solution:

Let $A = (1, 1, 1); B = (-2, 4, 1); C = (-1, 5, 5) \&$

$D = (2, 2, 5)$

then, $AB = \sqrt{9 + 9 + 0} = 3\sqrt{2}, BC = \sqrt{1 + 1 + 16} = 3\sqrt{2}$

and $CD = 3\sqrt{2}$ **and** $AD = 3\sqrt{2}$. **Since, all sides are equal. Hence it is a square.**

Q.11 Answer:

-1

Solution:

Suppose P divides QR in ratio $\lambda : 1$. **Then, co-ordinates of** P **are** $\left(\frac{5\lambda+2}{\lambda+1}, \frac{\lambda+2}{\lambda+1}, \frac{-2\lambda+1}{\lambda+1}\right)$

It is given that the x **-coordinate of** P **is 4 .**

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i.e., $\frac{5\lambda+2}{\lambda+1}=4\Rightarrow\lambda=2$

So, z **-coordinate of** P **is** $\frac{-2\lambda+1}{\lambda+1}=\frac{-4+1}{2+1}=-1.$

Q.12 Answer:

7

Solution:

The length of the edges are given by $a=5-2=3, b=9-3=6, c=7-5=2$**, so length of the diagonal**

$=\sqrt{a^2+b^2+c^2}=\sqrt{9+36+4}=7$ **units**

Q.13 Answer:

$-8x+4y-16z-16=k$

Solution:

$\because PA^2-PB^2=k$

$\therefore \left[(x-2)^2+(y-3)^2+(z-4)^2\right]$

$-\left[(x+2)^2+(y-5)^2+(z+4)^2\right]=k$

or $-8x+4y-16z-16=k$

Q.14 Answer:

2 and 3

Solution:

c), $Q(a+2, b+2, c-2)$

Given that $\mathrm{P}(\mathrm{a}, \mathrm{b},$ **and** $\mathrm{R}(\mathrm{a}+6,\ \mathrm{b}+6, \mathrm{c}-6)$ **are collinear, one point must divide, the other two points externally or internally. Let** R **divide** P **and** Q **in ratio** $k:1$ **so,**

taking on x **-coordinates** $\frac{k(a+2)+a}{k+1}=a+6$

$\Rightarrow \quad \mathrm{k}(\mathrm{a}+2)+\mathrm{a}=(\mathrm{k}+1)(\mathrm{a}+6)$

$\Rightarrow \quad ka+2k+a=ka+6k+a+6\Rightarrow -4k=6$ **or**

$k=-\frac{3}{2}$

Negative sign shows that this is external division in ratio $3:2$**. So,** R **is divided** P **and** Q **externally in** $3:2$ **ratio. Putting this value for** $y-$ **and** z **- coordinates satisfied :**

for y **- cordinate :**

$\frac{3(b+2)-2b}{3-2}=3b+6-2b=b+6$

and for z-cordinate:

$\frac{3(c-2)-2c}{3-2}=\frac{3c-6-2c}{1}=c-b$

Statement (2) **is correct. Also, let Q divide** P **and** R **in ratio** $P:1$ **taking an x-co-ordinate :**

$\frac{p(a+6)+a}{p+1}=a+2$

$\Rightarrow \frac{p\cdot a+6p+a}{p+1}=a+2$

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$\Rightarrow \mathrm{pa}+6\mathrm{p}+\mathrm{a}=\mathrm{pa}+\mathrm{a}+2\mathrm{p}+2 \Rightarrow 4\mathrm{p}=2$

$\Rightarrow \mathrm{p}=\frac{1}{2}$

Positive sign shows but the division is internal and in the ratio $1:2$ Verifying for y -and z - coordinates, satisfies this results.

For y co-ordinate, $\frac{(b+6)\times 1+2b}{3}=\frac{3b+6}{3}$

$=b+2$

and for z -coordinate, $\frac{c-6+2c}{3}=\frac{3c-6}{3}=c-2$

values are satisfied. So, statement (3) is c

Q.15 Answer:

$x-2z=0$

Solution:

Let $(\mathrm{h},\mathrm{k},\ell)$ be the point which is equidistant from the points $(1,2,3)$ and $(3,2,-1)\Rightarrow$

$\sqrt{(\mathrm{h}-1)^2+(\mathrm{k}-2)^2+(\ell-3)^2}$

$=\sqrt{(\mathrm{h}-3)^2+(\mathrm{k}-2)^2+(\ell+1)^2}$

$\Rightarrow \quad (h-1)^2+(\ell-3)^2=(h-3)^2+(\ell+1)^2$

$\Rightarrow \mathrm{h}^2+1-2\,\mathrm{h}+\ell^2-6\ell+9=\mathrm{h}^2-6\,\mathrm{h}+9+$

$\ell^2+2\ell+1$

$\Rightarrow -2\,\mathrm{h}-6\ell=-6\,\mathrm{h}+2\ell \Rightarrow \quad 6\,\mathrm{h}-2\,\mathrm{h}-6\ell$

$-2\ell=0$

$\Rightarrow 4\,\mathrm{h}-8\ell=0 \Rightarrow \mathrm{h}-2\ell=0$

Putting $\mathrm{h}=\mathrm{x}$ and $\ell=\mathrm{z}$

We get locus of points $(\mathrm{h},\mathrm{k},\ell)$ as, $\mathrm{x}-2\mathrm{z}=0$

Q.16 Answer:

Any four vectors in 3 - d space are always linearly dependent

Solution:

Any four vectors in 3 - d **space are always linearly dependent.**

Q.17 Answer:

$-2:3$

Solution:

Let the ratio be $\lambda:1$

The dividing plane is YZ plane.

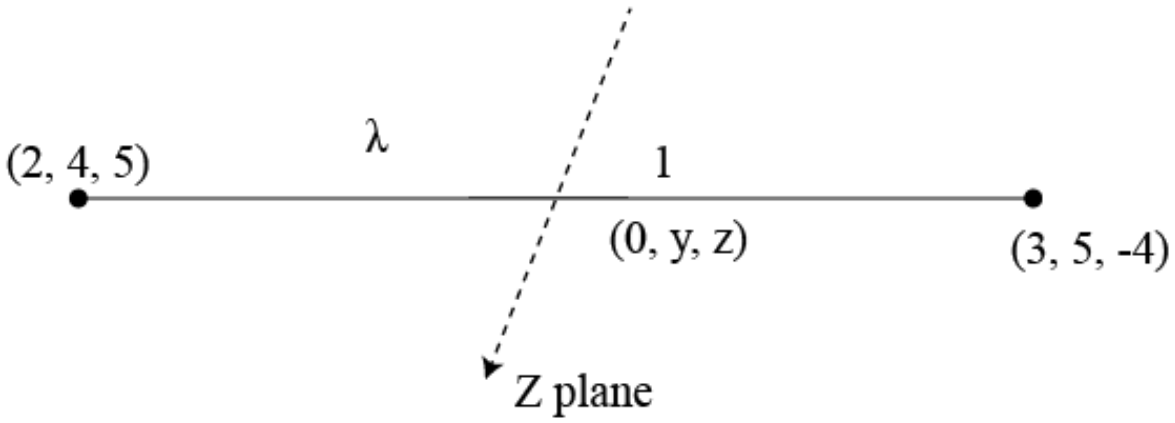
So, the co-ordinate of x after dividing the line $=0$

Now According to internal division rule of line segment $\mathrm{x}=\frac{3\times\lambda+1\times 2}{\lambda+1}\Rightarrow\frac{3\lambda+2}{\lambda+1}=0\Rightarrow 3\lambda+2=0\Rightarrow\lambda=\frac{-2}{3}$.

So. Ratio $=-2:3$

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i.e.



Q.18 Answer:

-2

Solution:

Let $A(5,2,4), B(6,-1,2), C(8,-7,k)$ **be the given points.**

Direction Ratios of AB **are** $< 6-5, -1-2, 2-4 >$ $i.e., < 1, -3, -2 >$

Direction Ration of BC **are** $< 8, -6, -7, +1k-2 >$

i.e., $< 2, -6, k-2 >$

Since A, B, C **are Collinear** $\therefore \frac{2}{1} = \frac{-6}{-3} = \frac{k-2}{-2}$

$\therefore k-2 = -4 \Rightarrow k = 2-4 = -2.$

Q.19 Answer:

$\left(\frac{\mathrm{a}}{2}, \frac{\mathrm{a}}{2}, \frac{\mathrm{a}}{2}\right)$

Solution:

Let the point $\mathrm{A}(\mathrm{x}, \mathrm{y}, \mathrm{z})$ **is equidistant from the points** $\mathrm{B}(\mathrm{a}, 0, 0), \mathrm{C}(0, \mathrm{a}, 0), \mathrm{D}(0, 0, \mathrm{a})$ **and** E $(0,0,0)$

Hence, $(x-a)^2 + y^2 + z^2 = x^2 + (y-a)^2 + z^2$

$= x^2 + y^2 + (z-a)^2$

$= \mathrm{x}^2 + \mathrm{y}^2 + \mathrm{z}^2$

$\Rightarrow \quad (x-a)^2 + y^2 + z^2 = x^2 + (y-a)^2 + z^2$

$\Rightarrow \quad x^2 + a^2 - 2ax + y^2 + z^2 = x^2 + y^2 + a^2 - 2ay + z^2$

$\Rightarrow \quad -2ax = -2ay \Rightarrow ax = ay \Rightarrow x = y$

Similarly, $\mathrm{ay} = \mathrm{az} \Rightarrow \mathrm{y} = \mathrm{z} \quad \Rightarrow \quad \mathrm{x} = \mathrm{y} = \mathrm{z}$

$\therefore \quad (\mathrm{x}-\mathrm{a})^2 + \mathrm{x}^2 + \mathrm{x}^2 = \mathrm{x}^2 + \mathrm{x}^2 + \mathrm{x}^2$

$\Rightarrow \quad x^2 + a^2 - 2ax + x^2 + x^2 = 3x^2$

$\Rightarrow \quad a^2 = 2ax \Rightarrow x = \frac{a}{2}$

$\therefore$ **Point is** $\left(\frac{\mathrm{a}}{2}, \frac{\mathrm{a}}{2}, \frac{\mathrm{a}}{2}\right).$

Q.20 Answer:

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externally in the ratio $2:3$

Solution:

externally in the ratio $2:3$

Let the XY - plane divide the line segment joining points $P(-1,3,4)$ and $Q(2,-5,6)$ in the ratio $\mathrm{k}:1$.

Using the section formula, the coordinates of the point of intersection are given by

$\left(\frac{k(2)-1}{k+1}, \frac{k(-5)+3}{k+1}, \frac{k(6)+4}{k+1}\right)$

On the XY - plane, the Z - coordinate of any point is zero.

$\Rightarrow \frac{k(6)+4}{k+1}=0$

$\Rightarrow 6k+4=0$

$\Rightarrow k=-\frac{2}{3}$

Thus, the XY - plane divides the line segment joining the given points in the ratio $2:3$ externally.

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